Loan Approval Prediction System – ER Diagram

Overview

The Entity Relationship (ER) Diagram represents the database structure of the **Loan Approval Prediction System**. It illustrates the main entities involved in the application, their attributes, primary keys, and the relationships between them. The ER diagram helps organize applicant information, loan requests, prediction results, machine learning model details, datasets, reports, and user sessions in a structured manner.

Description

The ER diagram provides a clear representation of how different components of the Loan Approval Prediction System interact with each other. It ensures efficient data management and supports the machine learning workflow used for loan approval prediction.

Entities

The system consists of the following primary entities:

- User
- Country
- HDI Input Data
- ML Model
- HDI Prediction
- Dataset
- Visualization Report
- Session

Primary Keys

Entity	Primary Key
User	user_id
Country	country_id
HDI Input Data	input_id
ML Model	model_id
HDI Prediction	prediction_id
Dataset	dataset_id
Visualization Report	report_id
Session	session_id

Relationships

- **User → HDI Input Data:** One user can submit multiple HDI prediction inputs (1:M).
- **Country → HDI Input Data:** One country can have multiple HDI input records (1:M).
- **HDI Input Data → HDI Prediction:** Each input record generates one prediction (1:1).
- **ML Model → HDI Prediction:** One machine learning model can generate multiple predictions (1:M).

- **HDI Prediction → Visualization Report:** One prediction can produce multiple visualization reports (1:M).
- **Dataset → ML Model:** One dataset can be used to train multiple machine learning models (1:M).
- **User → Session:** One user can have multiple application sessions (1:M).

User Entity Attributes

The **User** entity contains the following attributes:

- **user_id** (Primary Key)
- **name**
- **email**
- **role**
- **created_at**

Purpose

This ER diagram is designed to:

- Organize application data efficiently.
- Maintain relationships between users, datasets, and prediction results.
- Support machine learning model management.
- Store prediction outputs and visualization reports.
- Enable scalable database design for future enhancements.

Technologies Used

- ER Diagram Design
- Database Modeling
- Machine Learning Data Management
- GitHub Documentation

Outcome

The ER diagram provides a well-structured database design that supports the Loan Approval Prediction System by ensuring data consistency, reducing redundancy, and improving the management of prediction workflows.